MicroChipper

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In 2000, MicroChipper Corp (MCC), a manufacturer of custom logic devices, was considerdering In 2002 the MicroChipper Company (MCC) was considering the production of a logic device which would require 20 million transistors. Management was interested in whether it should be made as a single integrated device, or as several individual chips that would then be installed into a small printed circuit board.

In undertaking this project, the company was prepared to use 150 mm wafers and forecast a total cost per processed wafer (before testing) of $1550 (raw wafer cost, labor, supplies, equipment, and facilities).

The company planned on using the somewhat older 320 nm feature size technology, which was well understood. Cell areas factors, they estimated, would be comparable to those achieved in the logic portions of the Pentium 4. (See Exhibits). Silicon efficiency was projected to be 0.50.

With regard to other parameters, engineers estimated the defect density per cm as 0.35, the cluster parameter as 2, and the wafer exclusion band as 2 mm. They expected

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package cost per die to be $4 and final test cost to be $0.65 per die. Packaging yields had averaged 99% for the company and they expected final test yield to be 94%.

Management estimated the cost of inserting each chip into the final printed circuit board as $0.50. That cost included the “real estate” cost of the space taken on the board for a chip. The mechanical system to be replaced by the chip or chipset cost $75 per unit. Engi-neering design of the new chip or chipset cost about $100,000.

Should the device be built as a single integrated circuit or as several? What will be the average system cost? What volume is necessary to justify a movement to digital logic?

Exhibit 2
Integrated Circuit Cost Structures

		Year	1988	1998	2002	2002
		Product	DRAM	Pentium	DRAM	Pentium 4
			1 Mbit	II	256 Mbit	512KB
						Cache
A		Logic Transistors (millions)	0	7.5	0	41.9
B		Memory Transistors (millions)	1.024	0	256	13.1
C		Total Transistors (millions)	1.024	7.5	256	55
D		Wafer Diameter (mm)	150	150	200	200
E		Feature Size (nm)	1000	800	130	130
F		Wafer Starts/Week	5600	4500	6900	6900
G		Number of Masking Steps	10	17	25	23
		Cost of Processed Wafer ($/Wafer)				
H		Raw Wafer	40	60	60	85
J		Labor	132	163	102	174
K		Supplies	116	70	261	522
L		Equipment	132	497	962	1030
M		Facilities	12	38	93	89
N		*Total*	400	828	1478	1900
P		Die Size (sq-mm)	29	294	55.1	145.8
Q		Defect Density (per sq cm)	1.5	0.4	0.2	0.35
R		Cluster Parameter	3	1	2	1
S		Wafer Exclusion (mm)	2	2	3	3
T						
U	GrossDie()	Gross Die	551	48	510	185
V		Test/Control Sites	5	0	5	3
W	U-V	Gross Available Die	546	48	505	182
X	(1+P*Q/R)^-R	Probe Yield	67%	46%	90%	66%
Y	int(p*n)	Good Die	366	22	453	120
Z	N/Y	Cost/Good Die ($)	1.09	37.64	3.26	15.83
a		Package Cost ($/die)	0.15	2	0.6	8
b		Packaging Yield	0.98	0.99	0.99	0.99
c	int(a*b)	Good Packages	358	21	448	118
d		Final Test Cost ($/die)	0.16	0.16	0.15	0.5
e		Final Test Yield	0.9	0.95	0.99	0.95
f	int(c*e)	Good Chips	322	19	443	112
g	(N+Za+dc)/f	Cost/Good Chip ($)	1.59	46.07	4.10	26.06
h	g/B	Cost/Mbit ($)	1.59		0.016	
	g/C	Cost/M Transistors ($)	1.59	6.14	0.016	0.474

Notes to Exhibit 2

Costs have been estimated by combining technical information on devices with a number of industry sources, including Intel, Merrill Lynch, Goldman Sachs, IC Knowledge, SemaTech's "Wafercost Calculator," interviews with Mr. Howard Dicken, and the "Spreadsheet Model for Fab Economic Analysis," created by Leachman, Ding, and Sato-Misaw's UC Berkeley.

The yield equation used in Exhibit 2 was that given by :

$$Y = \left(1 + \frac{AD_d}{\alpha}\right)^{-\alpha}.$$

where the expected number of killer defects per mm^2 is D(the defect density), the active area of a die is A_d mm^2, and α is the "clustering" parameter, reflecting the degree to which particle positions were correlated. Normally, α had a value in the neighborhood of 2, but ranged from 0.3 to 5.

The DieCount() function calculates the number of die that fit on a wafer. This function is implemented in VBA as follows:

```
Function DieCount( height As Double, width As Double, diameter As Double) As Integer
' (c) 2003 Richard P. Rumelt
'This function computes the number of die that can be fit onto a wafer.  It takes the 'height and width of a
single die and the diameter of the wafer as inputs and returns an 'integer count.
Dim h As Double
Dim w As Double
Dim row As Integer
Dim columns As Integer
Dim rowmax As Integer
DieCount = 0
h = height
w = width
If h > w Then
 h = width
 w = height
End If
If w < diameter Then
 rowmax = Int(Sqr(diameter ^ 2 - w ^ 2) / h)
 For row = 1 To rowmax
      columns = Int(Sqr(diameter ^ 2 - (row * h) ^ 2) / w)
      DieCount = DieCount + columns
End If  Next row
End Function
```

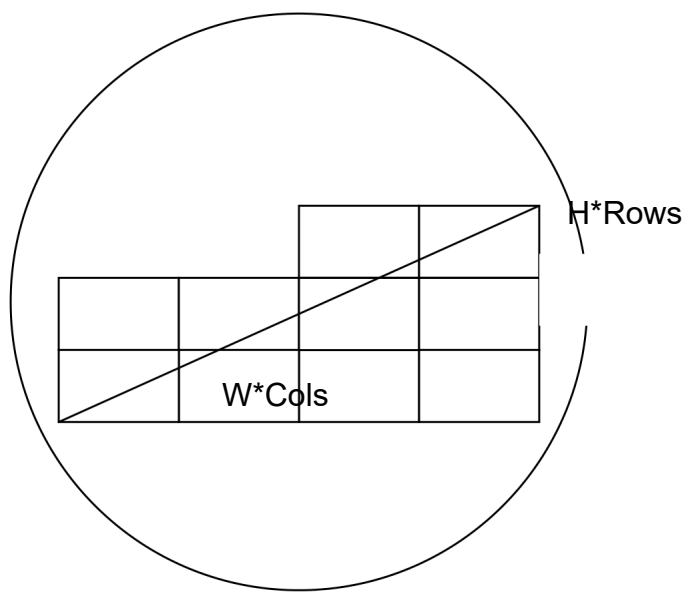

Exhibit 3
Die Area Computations

Year				2002	2002
Product				DRAM	Pentium 4
				256 Mbit	512 Kb Cache
Logic Transistor Area					
	Cell area factor			7	80
	Cell area (sq-microns)			0.1183	1.352
	Cells (million)			262	42.8
	Total Cell Area (sq-mm)			31.01	57.87
SRAM Transistor Area					
	Cell area factor				16.8
	Cell area (sq-microns)				1.352
	Cells (million)				12.2
	Total Cell Area (sq-mm)				16.49
Total Active Cell Area (Sq-mm)					74.36
Silicon Efficiency				0.55	0.51
Die Area (sq-mm)				56.38	145.80

Notes to Exhibit 3

Cell area factors were multipliers that related the square of feature size to the area taken up by a transistor. Thus, if the feature size was F microns, and the cell area factor was 20, then the area taken by one transistor was $20F^2$ square microns. There were 1 mil-lion square microns in 1 square millimeter, so 1 million transistors would take up $20F^2$ square millimeters.

The active cell area on an integrated circuit were surrounded by a relatively empty margin which housed the metal leads running out to the terminals on the edge of the chip. Often, one-half of the area of the chip was devoted to this margin. The proportion of the chip used for actual active devices was termed the *silicon efficiency*.

DRAM memory normally used 1 transistor per bit. The actual data was stored as a charge on a capacitor. SRAM memory was much faster, but required 3 transistors per bit of storage. The fast cache memory included on microprocessors and logic chips was generally SRAM. A 512KB L2 cache required 0.512*8*3 = 12.3 million transistors. The 13.056 transistor count for SRAM on the P4 (above) included smaller caches as well, but excluded the instruction pipeline storage on the chip.